TEAM AQUA GUARDIANS

AQUA GUARDIAN ROBOT

An Autonomous Water-Cleaning & Automatic Fish-Feeding Robot



1. Abstract

The Aqua Guardian Robot is a low-cost floating robotic system designed to address two common challenges in ponds, lakes and aquaculture environments: the accumulation of floating waste and the need for regular fish feeding. The robot moves across the water surface while collecting floating waste and storing it for later disposal. At the same time, an automatic feeding mechanism releases a controlled amount of fish food at fixed time intervals, helping reduce manual effort and food wastage. By combining water-surface cleaning and automated fish feeding in one system, the project aims to support cleaner water bodies and better aquatic-life care. The design also provides scope for future improvements such as solar-powered operation, water-quality monitoring, mobile-app-based control, GPS navigation and AI-based obstacle avoidance.

2. Project Overview

Water pollution is not only a human problem — it is a serious threat to aquatic life as well. Ponds, lakes and other water bodies often accumulate plastic, wrappers and other floating waste, which degrades water quality and harms fish and other aquatic organisms. To address this, our team has designed the Aqua Guardian Robot — a floating robot that cleans the water surface while also automatically feeding fish, combining environmental cleanup with aquatic-life care in a single low-cost device.

3. Problem Statement

a. Floating waste such as plastic, wrappers and debris collects on ponds, lakes and other water bodies, degrading water quality.

b. Polluted water harms fish and other aquatic organisms, disrupting the aquatic ecosystem.

c. Manual cleaning of water bodies is labor-intensive, time-consuming and often irregular.

d. Fish in ponds and aquaculture centers are frequently fed manually and inconsistently, leading to overfeeding, underfeeding and food wastage.

e. There is a need for a low-cost, automated system that can clean the water surface and maintain a healthy feeding schedule for fish without constant human supervision

4. Proposed Solution

Floating & Mobile Design: The robot rests on floaters that keep it stable on the water surface, while DC motors and propellers allow it to move freely across the water.

Waste Collection: As it moves, the robot collects floating waste and stores it in a solid waste storage basket, which can be removed and emptied later.

Automatic Fish Feeding: A dedicated container holds fish food, and a servo motor releases a controlled amount of food at fixed time intervals.

Future-Ready: The system is designed so that it can be upgraded to run on solar power, making it more eco-friendly and sustainable.

5. How the System Works

The robot floats on the water surface using its floater base and is propelled by DC motors connected to propellers, allowing it to navigate across ponds, lakes or tanks. While moving, it collects floating waste such as plastic and wrappers, which is stored in an onboard waste storage basket for later disposal.

In parallel, a separate subsystem manages fish feeding: a servo motor is triggered at fixed time intervals to release a pre-set amount of food from the food container into the water. This ensures the fish receive regular, controlled feeding while the robot continues its cleaning operation, reducing both manual effort and food wastage.

6. Market Opportunity

Clean and healthy water bodies are essential for aquaculture, recreation and the environment, and India has a growing network of ponds, lakes, fish farms and aquaculture centers that require regular maintenance. Government programs promoting sustainable aquaculture and pollution control create favorable ground for low-cost automated solutions. The Aqua Guardian Robot can serve fish farms, aquaculture centers, park and recreational lakes, temple ponds, and educational institutions looking for affordable environmental-technology demonstrators. Its key advantage is combining two functions — water cleaning and automatic fish feeding — in a single affordable device, unlike most existing solutions that address only one of these needs.

7. Expected Outcome & Future Scope

The prototype is expected to autonomously collect floating waste from a water body while simultaneously feeding fish at fixed intervals, reducing manual effort and improving water quality. Future versions can include solar power for fully self-sustained operation, water quality sensors (pH, turbidity, dissolved oxygen), remote monitoring through a mobile app, GPS-based navigation for larger water bodies, and AI-based obstacle avoidance and route optimization.

a. Solar-powered operation for continuous, eco-friendly functioning

b. Water quality sensors for real-time pollution monitoring

c. Mobile app dashboard for remote monitoring and control

d. Automatic obstacle detection and navigation for larger ponds/lakes

e. Scalable fish-feeding schedules customized per species

8. Conclusion

The Aqua Guardian Robot offers a low-cost, easy-to-use solution that goes beyond conventional cleaning devices by also caring for aquatic life through automated feeding. By combining floating waste collection with a servo-controlled fish-feeding mechanism, Team Aqua Guardians demonstrates how simple robotics and automation can reduce human effort, save time, and support a cleaner and healthier aquatic ecosystem — reflecting our broader goal of using technology to protect the environment.